Interview Q&A: scikit-learn

1. What is the difference between GridSearchCV and RandomizedSearchCV in scikit-learn?

GridSearchCV exhaustively searches all possible combinations of hyperparameters, while RandomizedSearchCV randomly samples from a specified distribution to search for optimal hyperparameters.

2. What is the purpose of the `transform` method in scikit-learn's pipeline?

The `transform` method applies the learned transformations to a given dataset, returning the transformed data.

3. What is the purpose of the `ColumnTransformer` class in scikit-learn?

The `ColumnTransformer` class is used to transform a subset of columns in a dataset, allowing for feature selection and transformation on specific features.

4. What is the difference between `StandardScaler` and `MinMaxScaler` in scikit-learn?

The main difference lies in how they scale the data. `StandardScaler` subtracts the mean and divides by standard deviation, while `MinMaxScaler` scales to a common range (usually 0 to 1). This affects the distribution of the features.

5. What is the purpose of the `Pipeline` class in scikit-learn?

The `Pipeline` class is used to chain multiple estimators together, allowing for a more complex workflow and easier management of data transformations.

6. What is the purpose of the `KernelPCA` class in scikit-learn?

The `KernelPCA` class is used to apply kernel principal component analysis, which can be useful for high-dimensional data where PCA may not work well.

7. What is the purpose of the `RobustScaler` class in scikit-learn?

The `RobustScaler` class scales features using statistics that are robust to outliers, such as the median and interquartile range.

8. What is the purpose of the `MultiTaskLearner` class in scikit-learn?

The `MultiTaskLearner` class is used for multi-task learning, where a single model learns to predict multiple related tasks.

9. What is the purpose of the `OneVsRestClassifier` class in scikit-learn?

It allows for multi-class classification by training one-vs-rest classifiers for each class.

10. What is the difference between `KNeighborsClassifier` and `KNeighborsRegressor` in scikit-learn?

The main difference lies in their output: `KNeighborsClassifier` predicts class labels, while `KNeighborsRegressor` predicts continuous values.

11. What is the purpose of the `LabelEncoder` class in scikit-learn?

The `LabelEncoder` class is used to convert categorical labels into numerical labels.

12. What is the purpose of the `VarianceThreshold` class in scikit-learn?

The `VarianceThreshold` class is used to filter out low-variance features from a dataset, which can improve model performance by reducing noise and irrelevant information.

13. How does the `Pipeline` class handle multi-threading in scikit-learn?

The `Pipeline` class uses a thread-safe implementation to handle multi-threading, ensuring that each step is executed sequentially and safely.

14. What is the purpose of the `MultiClassSVC` class in scikit-learn?

The `MultiClassSVC` class is a support vector classifier for multi-class classification problems, which can handle both linear and non-linear kernels.

15. What is the difference between `LogisticRegression` and `SGDClassifier` in scikit-learn?

The main difference lies in their optimization algorithms, with LogisticRegression using a Newton-CG method and SGDClassifier using stochastic gradient descent.

16. What is the purpose of the `make_scorer` function in scikit-learn?

The `make_scorer` function creates a scorer object from a given scoring metric, allowing it to be used with cross-validation and model selection.

17. What is the purpose of the `FeatureUnion` class in scikit-learn?

The `FeatureUnion` class allows combining multiple feature transformers into a single transformer, enabling more complex data preprocessing pipelines.

18. What is the purpose of the `LabelBinarizer` class in scikit-learn?

The `LabelBinarizer` class is used to transform labels into binary vectors, which can be useful for multi-class classification problems.

19. What is the purpose of the `MultiOutputRegressor` class in scikit-learn?

The `MultiOutputRegressor` class is used to wrap a regressor object to make multi-output regression possible.

20. What is the purpose of the `MultiTaskLearner` class?

The `MultiTaskLearner` class is not a part of scikit-learn, it seems to be a custom or external implementation.

21. How does the `KMeans` algorithm handle outliers?

The `KMeans` algorithm is sensitive to outliers, as it uses the squared Euclidean distance metric. This can lead to poor clustering performance if there are many outliers in the data.